Faculty of Engineering & Technology

Department of Computer Science

COMP333 – DATABASE SYSTEMS

Final Project Report

Birzeit University Bookstore Management System

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Project Description:

The main goal of this project is to develop a ***Bookstore Management System*** that helps manage the process of selling and renting books to students and faculty members in an organized and efficient way. The system is designed to store and manage all important information related to the bookstore, including books, authors, customers, suppliers, rentals, purchases, and stock levels.

This system helps the bookstore staff perform their daily tasks more easily by keeping all data in one place. Instead of using manual records, the system allows employees to track available books, record sales and rentals, and monitor stock quantities accurately. This reduces errors and saves time during daily operations.

In addition, the system supports better inventory management by showing which books are available, which books have low stock, and which books need to be reordered from suppliers. It also helps in tracking customer activities such as book purchases and rentals, including rental return dates.

The system can generate useful reports that help the bookstore management understand business performance. These reports include information such as best-selling books, available stock by category, and monthly sales performance. Overall, the Bookstore Management System aims to improve efficiency, organization, and decision-making while providing a simple and user-friendly way to manage bookstore operations.

Main System Functionality:

The system allows employees to perform daily operations efficiently such as:

- *Recording book sales and rentals*
- *Managing book inventory*
- *Tracking book deliveries from suppliers*
- *Managing authors and book categories*
- *Viewing business performance through detailed reports*

Entity Sets:

Entity	Description	Primary Key
Book	A collection of all book entities, each representing an individual book with details such as title, category, price, and availability	Book_ID
Author	A collection of all author entities, where each author may be linked to one or more books in the system	Author_ID
Customer	A collection of customer entities representing students or staff who purchase or rent books from the bookstore	Customer_ID
Supplier	A collection of supplier entities that provide books to the bookstore, including company and contact information	Supplier_ID
Sale	A collection of sale transaction entities, each recording the sale of a specific book to a customer on a particular date	Sale_ID
Rental	A collection of rental transaction entities that record book rentals, including rental date, due date, and return date.	Rental_ID
Stock	A collection of inventory records that track the available quantity of each book in the bookstore.	Book_ID
Category	A collection of category entities used to classify books by subject such as IT, History, or Medicine.	Category_ID

Relationships:

- *Each category can have multiple books*
- *Each supplier can supply multiple books*
- *Each customer can perform multiple sales*
- *Each customer can rent multiple books*
- *Each book can appear in multiple sales*
- *Each book can appear in multiple rentals*
- *Each book has exactly one stock record*

Queries:

Book Queries

- ✓ Retrieve all available books sorted by title
- ✓ Retrieve books by specific category
- ✓ Retrieve authors of a selected book
- ✓ Retrieve books with low stock (less than 10 copies)
- ✓ Count the number of books in each category

Customer & Sales Queries

- ✓ Retrieve all customers sorted by name
- ✓ Retrieve total sales revenue for a specific month
- ✓ Retrieve the top 5 best-selling books
- ✓ Retrieve sales details for a specific customer
- ✓ Retrieve books purchased on a specific date

Rental Queries

- ✓ Retrieve all rental books that are overdue
- ✓ Retrieve all rented books with their return dates
- ✓ Count total number of active rentals
- ✓ Retrieve rental history for a specific customer
- ✓ Retrieve most rented categories

Supplier & Inventory Queries

- ✓ Retrieve all suppliers sorted by company name
- ✓ Retrieve books supplied by a specific supplier
- ✓ Retrieve supplier contact info for a selected book
- ✓ Retrieve stock quantity for each book
- ✓ Retrieve inventory value based on price × quantity

Normalization & Schema SQL:

Normalization was applied to the database design to achieve **Third Normal Form (3NF)** by separating the **Authors, Categories, and Suppliers** into independent tables in order to avoid data redundancy.

Author details were separated into a dedicated table to ensure that each author's information is stored only once. This design eliminates duplication of data and improves data consistency, which satisfies the requirements of 3NF.

1) Book Queries:

- Retrieve all available books sorted by title:

```
SELECT * FROM books

WHERE availability = 1

ORDER BY title ASC;
```

- Retrieve books by a specific category (e.g., 'IT'):

```
SELECT b.*

FROM books b

JOIN categories c ON b.category_id = c.category_id

WHERE c.category_name = 'IT';
```

- Retrieve authors of a selected book (by ID):

```
SELECT a.author_name

FROM authors a

JOIN book_authors ba ON a.author_id = ba.author_id

WHERE ba.book_id = 1;
```

- Retrieve books with low stock (less than 10 copies):

```
SELECT b.title, s.quantity

FROM books b

JOIN stock s ON b.book_id = s.book_id

WHERE s.quantity < 10;
```

- Count the number of books in each category:

```
SELECT c.category_name, COUNT(b.book_id) AS book_count
FROM categories c
LEFT JOIN books b ON c.category_id = b.category_id
GROUP BY c.category_id;
```

2) Customer & Sales Queries:

- Retrieve all customers sorted by name:

```
SELECT *
FROM customers
ORDER BY customer_name ASC;
```

- Retrieve total sales revenue for a specific month (e.g., January):

```
SELECT b.title, COUNT(s.sale_id) AS sales_count
FROM books b
JOIN sales s ON b.book_id = s.book_id
GROUP BY b.book_id
ORDER BY sales_count DESC
LIMIT 5;
```

- Retrieve the top 5 best-selling books:

```
SELECT b.title, COUNT(s.sale_id) AS sales_count
FROM books b
JOIN sales s ON b.book_id = s.book_id
GROUP BY b.book_id
ORDER BY sales_count DESC
LIMIT 5;
```

- Retrieve sales details for a specific customer (by ID):

```
SELECT s.*, b.title
FROM sales s
JOIN books b ON s.book_id = b.book_id
WHERE s.customer_id = 1;
```

- Retrieve books purchased on a specific date:

```
SELECT b.title
FROM books b
JOIN sales s ON b.book_id = s.book_id
WHERE DATE(s.sale_date) = '2024-01-15';
```

3) Rental Queries:

- Retrieve all rental books that are overdue:

```
SELECT b.title, c.customer_name, r.due_date
FROM rentals r
JOIN books b ON r.book_id = b.book_id
JOIN customers c ON r.customer_id = c.customer_id
WHERE r.return_date IS NULL
AND r.due_date < NOW();
```

- Retrieve all rented books with their return dates:

```
SELECT b.title, r.rental_date, r.return_date
FROM rentals r
JOIN books b ON r.book_id = b.book_id;
```

- Count total number of active rentals:

```
SELECT COUNT(*) AS active_rentals
FROM rentals
WHERE return_date IS NULL;
```

- Retrieve rental history for a specific customer:

```
SELECT b.title, r.rental_date, r.return_date
FROM rentals r
JOIN books b ON r.book_id = b.book_id
WHERE r.customer_id = 1;
```

- Retrieve the most rented categories:

```
SELECT c.category_name, COUNT(r.rental_id) AS rental_count
FROM categories c
JOIN books b ON c.category_id = b.category_id
JOIN rentals r ON b.book_id = r.book_id
GROUP BY c.category_id
ORDER BY rental_count DESC;
```

4) Supplier & Inventory Queries:

- Retrieve all suppliers sorted by company name:

```
SELECT *
FROM suppliers
ORDER BY company_name ASC;
```

- Retrieve books supplied by a specific supplier (by ID):

```
SELECT b.title
FROM books b
WHERE b.supplier_id = 1;
```

- Retrieve supplier contact information for a selected book:

```
SELECT s.company_name, s.supplier_email, s.supplier_phone
FROM suppliers s
JOIN books b ON s.supplier_id = b.supplier_id
WHERE b.book_id = 1;
```

- Retrieve stock quantity for each book:

```
SELECT b.title, s.quantity
FROM books b
JOIN stock s ON b.book_id = s.book_id;
```

- Retrieve inventory value based on price × quantity:

```
SELECT SUM(b.price * s.quantity) AS total_inventory_value
FROM books b
JOIN stock s ON b.book_id = s.book_id;
```

Figures from the Project:

This figure illustrates key screenshots from the Bookstore Management System. It includes the **Admin Dashboard**, which displays system reports such as low stock alerts, overdue rentals, books per category, and total sales revenue, as well as the **User Interface**, which shows the available books with their titles, categories, prices, and stock quantities. These interfaces demonstrate how the system supports both administrative monitoring and user interaction in an organized and user-friendly manner.

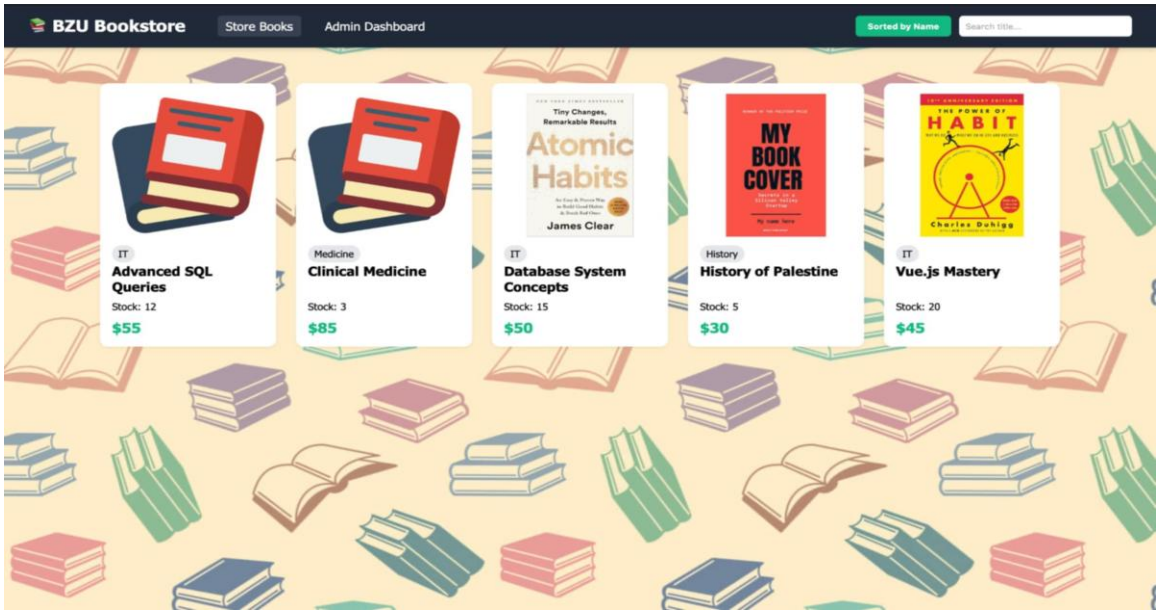

Figure 1

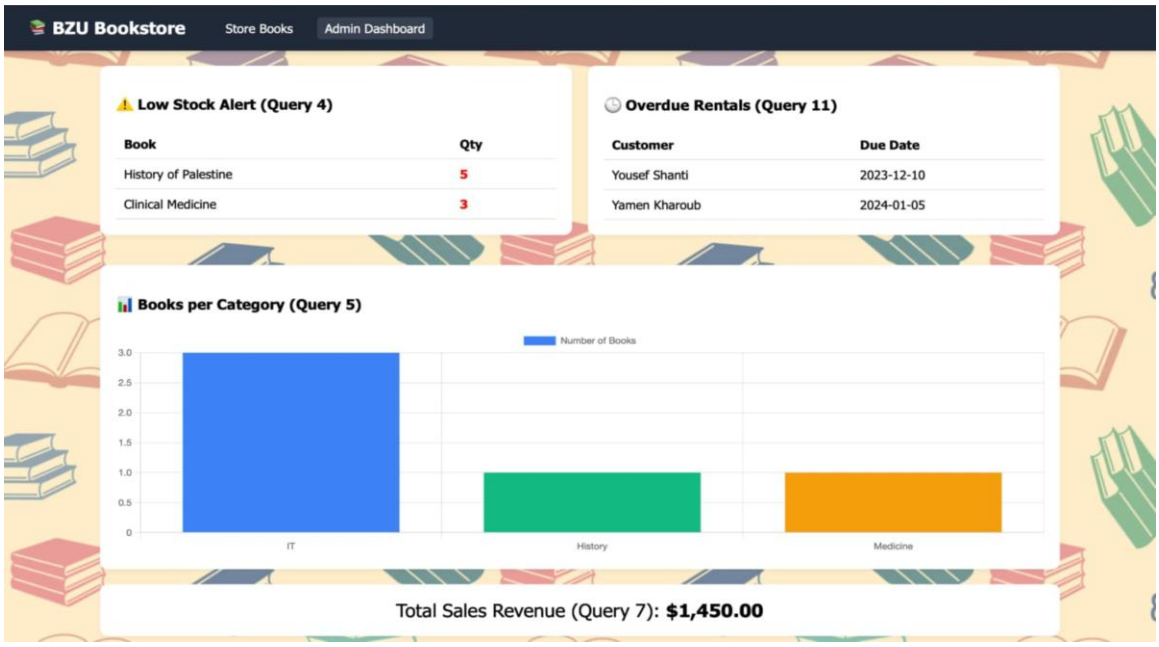

Figure 2